No matter how little, or how much mining hash power is applied, the average block is mined every 10 minutes.

- When the total **hash rate** decreases, the **difficulty level** decreases to make it easier for miners to mine new bitcoin.
    - This helps to keep steady the rate at which new ***bitcoin*** are mined.

- The difficulty adjustment is done using a formula that takes into account the average time it took to mine the previous 2016 blocks.
    - If the average time to mine 2016 blocks takes less than 14 days, the difficulty level is increased.
    - If the average time to complete 2016 blocks takes more than 14 days, the difficulty level is decreased.

The Issuance Rate refers to the rate at which new coins are being added to the circulating supply, often through mining. This rate can be impacted by various factors, including changes in network hash rate, the number of blocks mined, and halving events, which reduce the number of coins that can be mined per block.

8.5 The Step-by-Step Process of Mining a Block

Mining a block in the ***Bitcoin Network*** involves several steps:

1. New transactions are broadcast to the network, picked up, and verified by the nodes.
2. Transactions are collected from the unconfirmed transactions in the **Mempool**. Transactions with higher fees are prioritized.
3. These transactions are then organized in a **Merkle Tree** and included in a **candidate block**, along with the **previous Block's Hash**, a **timestamp**, and a **nonce**.
4. Miners compete to solve a mathematical puzzle based on the information in the block, including the transactions and a random number.
5. The puzzle involves finding a specific number (the "hash") that, when combined with the block data, results in a value that is less than a target number.

In other words, miners use **proof-of-work** to adjust the **nonce** value until a hash is obtained that meets the **specified difficulty** requirements.